Far-End Crosstalk Cancellation With MIMO OFDM for >200 Gb/s ADC-Based Serial Links

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***Abstract*—This brief presents an area- and power-efficient far-end crosstalk (FEXT) cancellation scheme with multi-input multi-output (MIMO) orthogonal frequency-division multiplexing (OFDM) for an analog-to-digital converter (ADC)-based ultra-high-speed serial links. The frequency-domain per-sub-channel equalization nature of OFDM allows a simple FEXT cancellation with MIMO equalizer in its digital signal processor (DSP), unlike time-domain pulse amplitude modulation (PAM) that requires heavy parallel cross-lane filtering for FEXT cancellation. With architectural modeling of a 2 × 2 MIMO OFDM wireline transceiver (TRX), simulation results show that the main and FEXT channels' frequency responses are properly estimated using pilot symbols. With the estimated channel responses, simulation results also show that the FEXT can be effectively canceled out with per-sub-channel MIMO equalization, achieving an overall symbol error rate (SER) of 1.4×10^{-5} at 224 Gb/s/lane for a channel having 23 dB loss at 40 GHz, while the SER is 4×10^{-1} without FEXT cancellation.**

***Index Terms*—Serial link, wireline transceiver, ADC-based serial link, orthogonal frequency division multiplexing, OFDM, wireline communications, wireline receiver, far-end crosstalk, FEXT.**

I. Introduction

THE INCREASING demands for ever-higher communication bandwidth for cloud computing platforms and data centers have been leading the per-lane data rate (*DR*) increase of wireline transceivers (TRX) from 28 Gb/s [1] to 112 Gb/s [2], [3] for the last decade. While non-return-to-zero (NRZ) was the dominating modulation for $DR < 56$ Gb/s with analog building blocks for its TRX, 4-level pulse amplitude modulation (PAM-4) with a high-speed analog-to-digital converter (ADC) and digital signal processor (DSP)-based receiver (RX) became popular for $DR \geq 56$ Gb/s [4], [5], [6], [7], [8]. Digital-to-analog converter (DAC)-based transmitter (TX) is widely used for $DR \geq 112$ Gb/s together with the ADC-based RX providing high flexibility and strong equalization capabilities [9], [10]. While the PAM-4 TRX with a DAC-based TX and ADC-based RX has kept pushing the per-lane communication speed, its extension to > 200 Gb/s for long-reach channels (> 35 dB of loss at Nyquist) is unclear. Moreover, handling the increasing far-end crosstalk (FEXT) typically relies on keeping a sufficient link margin, instead of canceling the FEXT [2].

For single-ended dense interconnects at a relatively low speed (7 Gb/s/lane), analog cross-equalizers including both continuous-time and decision-feedback equalizers between lanes are introduced in [11] for FEXT cancellation. However, analog cross-equalizers have limitations for being employed for $DR > 200$ Gb/s due to the bandwidth requirements and stringent timing constraints, and ADC-based PAM-4 cross equalizers will be excessively heavy in terms of power and area due to the required parallel long-tap finite impulse response (FIR) filters.

Employing orthogonal frequency-division multiplexing (OFDM) which is a frequency-domain modulation, time-domain convolution (filtering) operation reduces to element-wise multiplication for the equalization process, simplifying the RX equalizer DSP [12]. Demonstrated with a 56 Gb/s silicon prototype [12], extending OFDM for > 100 Gb/s serial links has been studied in [13], [14]. Benefiting from the per-sub-channel frequency-domain equalization nature of OFDM, this brief presents an area- and power-efficient FEXT cancellation for ultra-high-speed ADC-based serial links with multi-input multi-output (MIMO) OFDM RX equalizer.

The rest of this brief is organized as follows. The overall architecture of a wireline OFDM TRX is briefly reviewed in Section II. Section III explains the FEXT cancellation scheme with a 2 × 2 MIMO OFDM and the MIMO equalizer coefficient calculation. Section IV explains the 2 × 2 MIMO channel estimation flow. Simulation results are shown in Section V, and Section VI concludes this brief.

II. OFDM Wireline Transceiver and Frequency-Domain Equalization

A. Overall Data-Path

Fig. 1 shows the overall block diagram of a wireline TRX with OFDM modulation. On the TX side, data in the binary bitstream with forward error correction (FEC) encoding are mapped to the frequency-domain quadrature amplitude modulation (QAM) symbols. Optionally, bit-loading and/or power-loading can be applied to the frequency-domain QAM symbols before the symbols are transformed to the time-domain OFDM symbol by an inverse discrete Fourier transform (IDFT) engine. In the time domain, cyclic prefix

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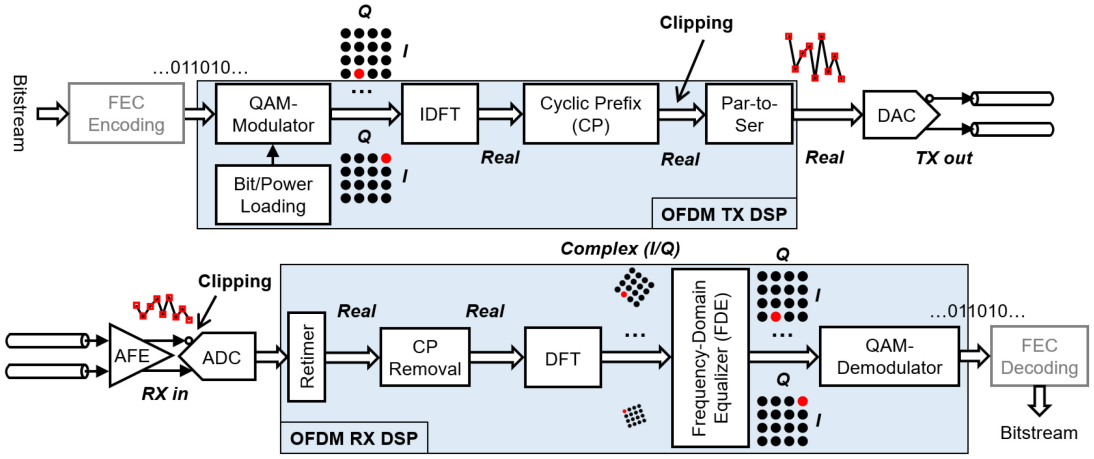


Fig. 1. Overall block diagram of the data-path of an OFDM TRX for a wireline serial link.

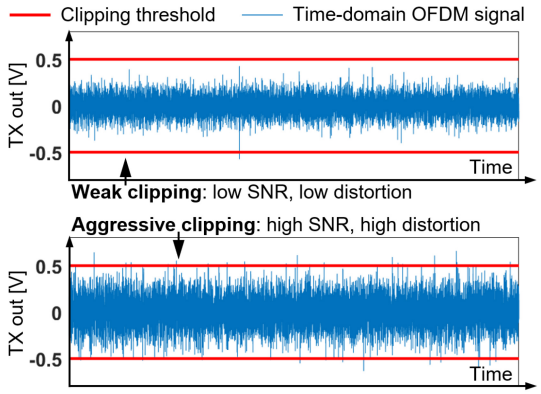


Fig. 2. Clipping of the time-domain OFDM signal on the TX: weak clipping does not cause a lot of distortion but the overall signal level is low, and aggressive clipping increases the overall signal power at the cost of increased distortion.

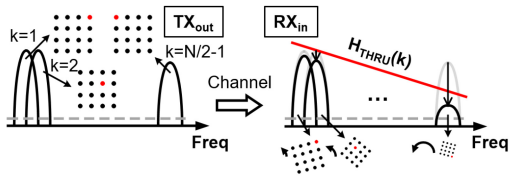


Fig. 3. Frequency-dependent amplitude attenuation and angular rotation of QAM symbols due to the channel.

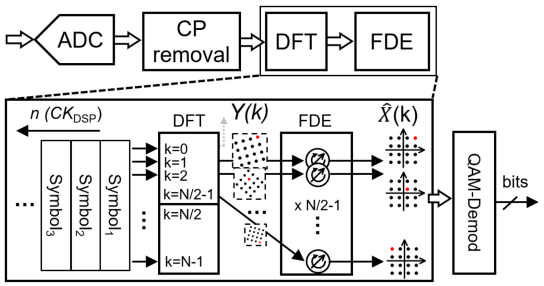


Fig. 4. Frequency-domain equalization by per-sub-channel amplitude adjustment and angular rotation correction.

(CP) is added to the OFDM symbol by simply copying the last *Ncp* samples in an OFDM symbol to the beginning of the OFDM symbol to protect the OFDM symbol from the inter-symbol interference (ISI) [15]. The CP-added OFDM symbol is then serialized for being transmitted toward the channel by a DAC. Clipping can be applied at this stage to trade off the distortion and the signal-to-noise ratio (SNR) [13], [16], as shown in Fig. 2.

On the RX side, the overall signal processing is the reverse operations of the TX-side signal processing. The analog front-end (AFE) provides high-frequency peaking with continuous-time linear equalizer (CTLE) for partial channel loss compensation and the overall signal gain with variable gain amplifier (VGA) in such a way that the overall signal amplitudes fit well with the ADC analog input dynamic range. Similar to the DAC clipping, the incoming OFDM signal is clipped at the ADC with a proper clipping ratio to balance the overall signal SNR and the clipping-induced distortion [13], [16]. The data sampled by a time-interleaved ADC is aligned by the retimer to be synchronized to the DSP clock, and the CP is removed from every OFDM symbol. The CP-removed time-domain OFDM symbol is transformed to the frequency domain by a discrete Fourier transform (DFT) engine.

QAM symbols appear again at this stage, with the noisy constellations and sub-channel-dependent amplitude reduction with angular rotation (Fig. 3) caused by the frequency-selective loss and phase delay of the channel. Frequency-domain equalizer (FDE) corrects the amplitudes and angles of the QAM symbols using a single complex multiplier per sub-channel. The equalized QAM symbols are then demodulated to the binary bitstream.

B. Frequency-Domain Equalization

One of the main advantages of using OFDM modulation for high-speed serial links is its per-sub-channel equalization nature. As shown in Fig. 4, the attenuated and rotated frequency-domain QAM symbol ($Y(k)$) on each sub-channel is equalized to $\hat{X}(k)$ by a single complex multiplier as

$$\hat{X}(k) = Y(k) \cdot H^{-1}(k), \tag{1}$$

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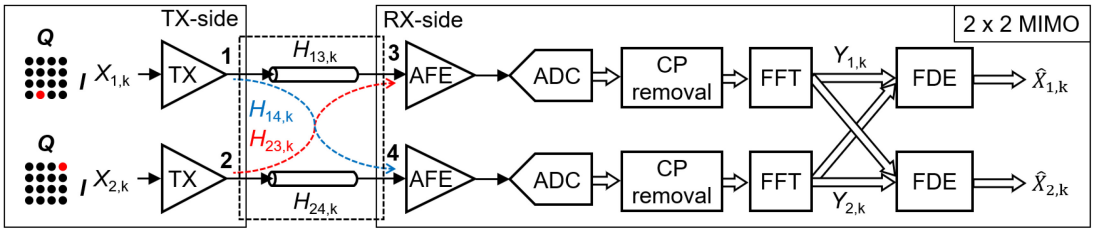


Fig. 5. Simplified block diagram of a 2 × 2 MIMO OFDM TRX for FEXT cancellation.

where $H^{-1}(k)$ is the inverse of the channel frequency response for k^{th} sub-channel, and $k \in \{1, \ldots, \frac{NFFT}{2} - 1\}$ is the sub-channel index with $NFFT$ being the length of the DFT and IDFT. It is worth noting that only the first half of the total DFT output bins are to be equalized (excluding the DC bin) since the second half is conjugate symmetry of the first half, containing the same information due to the Hermitian symmetry [12].

Such per-sub-channel equalization allows much simpler equalizer DSP implementation as compared to that of PAM-4 or PAM-8 which requires heavy feed-forward equalizer (FFE) based on parallel FIR filters. The number of taps for FFE reaches around 30 for 112 Gb/s data rate in addition to a speculative 1-tap DFE [2]. Considering a 64-way parallel data path, the number of real multipliers used for FFE is nearly 2K, while only around 100 real multipliers are needed for OFDM FDE with the same target data rate, excluding the resources related to the fast Fourier transform (FFT) that efficiently implements the DFT.

III. FEXT Cancellation With MIMO OFDM TRX

The single-lane OFDM serial link shown in Fig. 1 can be extended to a MIMO structure for FEXT cancellation. A simplified block diagram of a 2 × 2 MIMO OFDM TRX is shown in Fig. 5 considering the simplest FEXT scenario with only one aggressor. Note that the analog data paths in Fig. 5 are shown with single-ended drawing for the sake of simplicity, while they are actually differential. We assume that the single-lane SNR without the FEXT aggressor for every sub-channel k is sufficiently high to meet the desired target pre-FEC bit error rate (BER), e.g., $< 10^{-4}$. For an OFDM symbol transmission and reception, the received QAM symbols on k^{th} sub-channel $Y_{1,\text{k}}$ and $Y_{2,\text{k}}$ before the equalization can then be expressed as

$$\begin{pmatrix} Y_{1,\text{k}} \\ Y_{2,\text{k}} \end{pmatrix} = \begin{pmatrix} H_{13,\text{k}}X_{1,\text{k}} + H_{23,\text{k}}X_{2,\text{k}} \\ H_{14,\text{k}}X_{1,\text{k}} + H_{24,\text{k}}X_{2,\text{k}} \end{pmatrix}, \tag{2}$$

where $H_{13,\text{k}}, H_{24,\text{k}}$ and $H_{23,\text{k}}, H_{14,\text{k}}$ are the main and FEXT channel frequency responses at k^{th} sub-channel, respectively, and $X_{1,\text{k}}, X_{2,\text{k}}$ are the transmitted QAM symbols on k^{th} sub-channel in the frequency domain, for the 2 × 2 MIMO TRX shown in Fig. 5. Then, the estimates of the transmitted symbols $\hat{X}_{1,\text{k}}$ and $\hat{X}_{2,\text{k}}$ can be obtained by

$$\begin{pmatrix} \hat{X}_{1,\text{k}} \\ \hat{X}_{2,\text{k}} \end{pmatrix} = \mathbf{H}_\text{k}^{-1} \begin{pmatrix} Y_{1,\text{k}} \\ Y_{2,\text{k}} \end{pmatrix} \tag{3}$$

with

$$\mathbf{H}_\text{k} = \begin{pmatrix} H_{13,\text{k}} & H_{23,\text{k}} \\ H_{14,\text{k}} & H_{24,\text{k}} \end{pmatrix}. \tag{4}$$

Rewriting (3) with $\mathbf{P}_\text{k} = \mathbf{H}_\text{k}^{-1}$,

$$\begin{pmatrix} \hat{X}_{1,\text{k}} \\ \hat{X}_{2,\text{k}} \end{pmatrix} = \begin{pmatrix} P_{1,\text{k}}Y_{1,\text{k}} + P_{2,\text{k}}Y_{2,\text{k}} \\ P_{3,\text{k}}Y_{1,\text{k}} + P_{4,\text{k}}Y_{2,\text{k}} \end{pmatrix} \tag{5}$$

with

$$\mathbf{P}_\text{k} = \begin{pmatrix} P_{1,\text{k}} & P_{2,\text{k}} \\ P_{3,\text{k}} & P_{4,\text{k}} \end{pmatrix} = \frac{1}{\det(\mathbf{H}_\text{k})} \begin{pmatrix} H_{24,\text{k}} & -H_{23,\text{k}} \\ -H_{14,\text{k}} & H_{13,\text{k}} \end{pmatrix}. \tag{6}$$

It can be seen from (5) that, taking the FEXT into account, the estimated symbol $\hat{X}_\text{k}$ for each lane is obtained using one additional complex multiplication (and an adder) in addition to the simple single-lane FDE shown in Fig. 4. It is worth noting that the MIMO equalization parameters for each sub-channel k, $\mathbf{P}_\text{k}$, can be estimated relatively slowly in source-synchronized wireline communications, e.g., once a millisecond, as the channels are slowly varying. Hence, circuit-level hardware overhead coming from the channel estimation to obtain $\mathbf{P}_\text{k}$ is negligible.

IV. 2 × 2 MIMO Channel Estimation

After the link power-up, the OFDM TX first sends a synchronization sequence so that the RX can find the CP index. Once the TX and the RX are synchronized, pilot symbols are sent for the RX channel estimation (CHEST). The CHEST is repeated every transmission of M OFDM symbols; hence the effective data rate degradation is $100 \times \frac{1}{M}\%$, which is negligible for a sufficiently large M such as $M > 100$.

A. Equalization Coefficient Calculation

For a 2 × 2 MIMO CHEST, two OFDM symbols are transmitted on each TRX lane at the same time period as described below. Firstly, for $n = 1$ where $n \in \{1, 2\}$ is the OFDM symbol index, two OFDM symbols with QAMs having the same real and imaginary magnitudes but with opposite polarities ($X_{1,\text{k}} = S_\text{k}$ and $X_{2,\text{k}} = -S_\text{k}$) for each sub-channel are transmitted through the two TRX lanes, respectively. Then, the received QAM symbols for each sub-channel on the two lanes can be expressed as

$$Y_{1,\text{k}}^{(\text{n}=1)} = H_{13,\text{k}}S_\text{k} - H_{23,\text{k}}S_\text{k} \tag{7}$$

$$Y_{2,\text{k}}^{(\text{n}=1)} = H_{14,\text{k}}S_\text{k} - H_{24,\text{k}}S_\text{k}, \tag{8}$$

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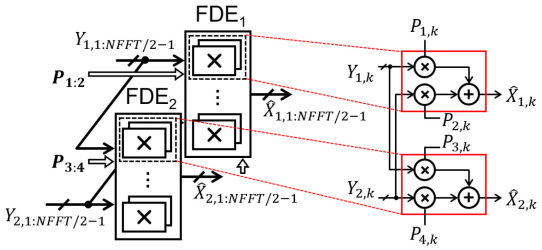


Fig. 6. Circuit implementation example of FDE for 2 × 2 MIMO FEXT cancellation and the unit FDE cell for each sub-channel k (inside the red rectangle).

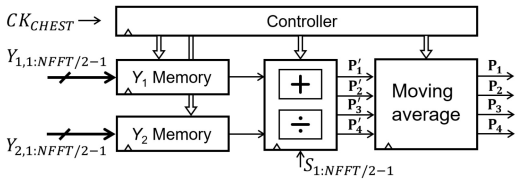


Fig. 7. Block diagram of the CHEST circuit. The throughput of the CHEST can be low as the source-synchronized wireline channels slowly vary.

TABLE I
COMPARISON OF THE DSP EQUALIZERS FOR OFDM AND ADC-BASED PAM-4

	Equalization domain	Equalization type	FEXT cancellation	Area in #(multipliers)
MIMO OFDM	Frequency	FFT+FDE	Yes	Hundreds*
ADC-based PAM-4	Time	FFE+DFE***	No	Thousands**

*With partially-serial FFT shown in [16] and a 64-way time-interleaved ADC.
**For an FFE having more than 31 taps with a 64-way time-interleaved ADC.
***Decision feedback equalizer

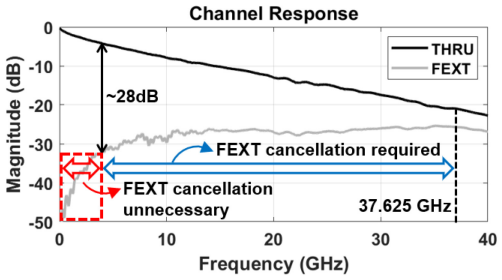


Fig. 8. Measured channel frequency response used for simulations, where the necessity of FEXT cancellation depends on the loss difference between THRU and FEXT responses.

where $Y_{m,k}^{(n=1)}$ is the received channel estimation symbol on the kth sub-channel when $n = 1$ with $m \in \{1, 2\}$ being the lane index. Secondly, for $n = 2$, two identical OFDM symbols S_k for each sub-channel k are transmitted through the two lanes. Hence, the received symbols are expressed as

$$Y_{1,k}^{(n=2)} = H_{13,k}S_k + H_{23,k}S_k \quad (9)$$

$$Y_{2,k}^{(n=2)} = H_{14,k}S_k + H_{24,k}S_k. \quad (10)$$

By subtracting (7) from (9) and by adding (7) and (9), we obtain $H_{23,k}$ and $H_{13,k}$, respectively, as

$$H_{23,k} = \frac{Y_{1,k}^{(n=2)} - Y_{1,k}^{(n=1)}}{2S_k}, \quad H_{13,k} = \frac{Y_{1,k}^{(n=2)} + Y_{1,k}^{(n=1)}}{2S_k}. \quad (11)$$

Similarly, by subtracting (8) from (10) and by adding (8) and (10), we obtain H_{14} and H_{24}, respectively, as

$$H_{14,k} = \frac{Y_{2,k}^{(n=2)} + Y_{2,k}^{(n=1)}}{2S_k}, \quad H_{24,k} = \frac{Y_{2,k}^{(n=2)} - Y_{2,k}^{(n=1)}}{2S_k}. \quad (12)$$

Since $Y_{1,k}^{(n=1)}, Y_{1,k}^{(n=2)}, Y_{2,k}^{(n=1)}, Y_{2,k}^{(n=2)}$ are measured values on the RX and S_k is a known value, with two OFDM symbol transmission and reception, $\mathbf{H}_k$ for all k can be obtained, and therefore, $\mathbf{P}_k$ can be calculated. Having $\mathbf{P}_k$, the equalized symbols $\hat{\mathbf{X}}_k$ with FEXT cancellation can be calculated with (5).

With the presence of random noise caused by several sources such as DAC and ADC quantization, thermal noise, AFE non-linearity and so on, the $\mathbf{P}_k$ can be moving-averaged for more accurate CHEST. Hence, after the link power-up, the CHEST phase can include hundreds of pilot symbols ($< 10\,\mu s$ in time with $NFFT = 1024$ and $DR = 224$ Gb/s) for an accurate CHEST, and then slowly moving-averaged to track the channel characteristic variation.

B. 2 × 2 FDE Circuit

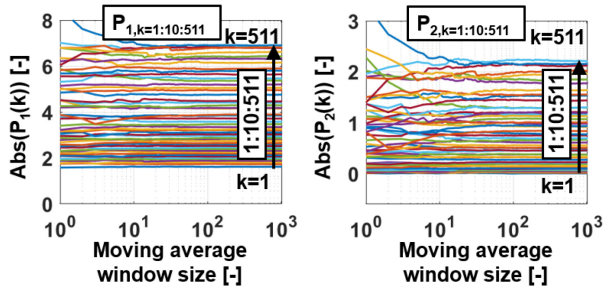
The structure of the 2×2 MIMO FDE is shown in Fig. 6. By having only one more complex multiplier for each sub-channel equalizer on top of the single-lane FDE, the attenuated/rotated QAM symbol on each DFT output bin can be equalized with FEXT cancellation. The unit FDE cells shown inside the red rectangles in Fig. 6 perform (5) with properly estimated $P_{1:4,k}$ for k^{th} sub-channel of the two RX lanes. The $\mathbf{P}_k$ is estimated by a CHEST circuit of which block diagram is shown in Fig. 7. Once the un-equalized frequency-domain QAM symbols on the two RX lanes ($Y_{1,1:NFFT/2-1}$, $Y_{2,1:NFFT/2-1}$) are captured, the CHEST engine performs (7)–(12) with simple arithmetic followed by 2×2 matrix inversions for obtaining $\mathbf{P}'_{1:4}$. As $\mathbf{P}'_{1:4}$ are inaccurate due to random noise, moving average is applied with sufficiently large averaging window L (e.g., $L > 100$) to obtain $\mathbf{P}_{1:4}$.

V. SIMULATION RESULTS

For simulations, a two-lane OFDM TRX is modeled with $NFFT = 1024$, $Ncp = 8$, 64-QAM symbols for all sub-channels, data rate $DR = 224$ Gb/s, OFDM-occupied bandwidth of 37.625 GHz, DAC/ADC sampling rate $F_s = 75.25$ GS/s, 7-bit DAC/ADC, 1 % UI root-mean-square (RMS) TX/RX sampling jitter, 1.8 mV RMS RX input random noise, 1 $V_{pp,diff}$ DAC output full-scale, and 0.5 $V_{pp,diff}$ ADC input full-scale. The clipping threshold-to-average magnitude ratios are 12.6 dB on the TX output and 13.4 dB on the RX input. The CTLE is not considered and the RX VGA amplifies the input signal so that the ADC uses its input full-scale to sample the input signal. The OFDM TX applies -10.5 dB of de-emphasis at sub-channel $k = 1$, and fewer de-emphasis levels that are linearly proportional to the sub-channel index k in dB-scale in such a way that the gain is 0 dB at $k = 511$.

The measured channel frequency response [17] used for the simulations is shown in Fig. 8. It can be seen from the highlighted area by a dotted red rectangle in Fig. 8 that the FEXT frequency response is weak as compared to the THRU response at low-frequency, exhibiting about 28 dB of main-to-FEXT ratio at sub-channel $k = 50$ corresponding to the

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(a) (b)

Fig. 9. Magnitudes of the $P_{1,k}$ (a) and $P_{2,k}$ (b) for $k \in \{1, 11, \ldots, 511\}$ with respect to the moving average window size.

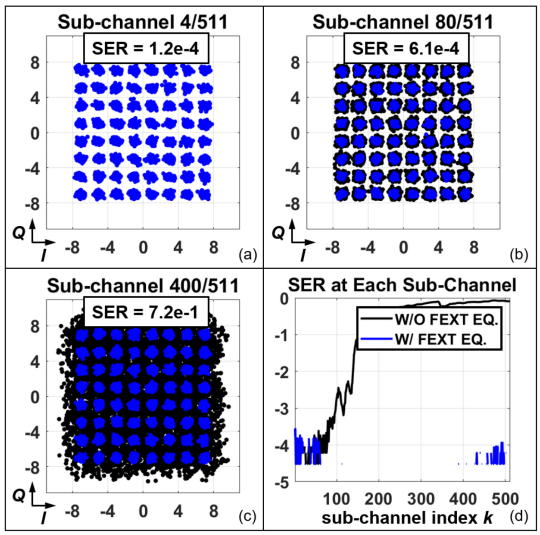


Fig. 10. Constellation diagrams for selected sub-channels with (blue markers) and without (black markers) FEXT cancellation: a low-frequency sub-channel where FEXT is totally negligible (a), a mid-low-frequency sub-channel where FEXT starts to affect the sub-channel SER (b), a high-frequency sub-channel where FEXT is significant (c), and SER at each sub-channel (d).

frequency of about 3.7 GHz given simulation settings. Fig. 9(a) and Fig. 9(b) show the magnitudes of $P_{1,k}$ and $P_{2,k}$, respectively, with respect to the moving average window size L from 1 to 1000, for $k \in \{1, 11, \ldots, 511\}$, while displaying for all $k \in \{1, 2, \ldots, 511\}$ will result in too crowded figures. As L increases, $P_{1,k}$ and $P_{2,k}$ become less noisy due to the noise averaging, and from $L \geq 100$, they are well stabilized. The convergence trend for $P_{3,k}$ and $P_{4,k}$ is the same as $P_{1,k}$ and $P_{2,k}$ while they are not plotted in this brief.

Constellation diagrams after FDE without (black markers) and with (blue markers) are shown in Fig. 10(a)–(c). The SER at each sub-channel without and with FEXT cancellation is plotted in Fig. 10(d): it can be seen that the low-frequency sub-channels (up to $k \approx 50$) are practically not affected by FEXT, while higher frequency sub-channels clearly benefit from the FEXT cancellation. The overall SER without and with FEXT cancellation is 4×10^{-1} and 1.4×10^{-5}, respectively, and the RX with FEXT cancellation achieves pre-FEC BER of 2.4×10^{-6} considering gray coding. Table I provides qualitative comparisons between the DSPs with PAM-4 and MIMO OFDM for communicating at 224 Gb/s of data rate through channels with $>$ 32 dB loss at 56 GHz.

VI. Conclusion

Area- and energy-efficient FEXT cancellation scheme with MIMO OFDM for ultra-high-speed electrical serial links is presented. The MIMO OFDM TRX with the frequency-domain per-sub-channel equalization nature effectively cancels out the FEXT with a lightweight DSP overhead. The cost to pay is only a single complex multiplier per sub-channel per aggressor, while time-domain modulations such as PAM-4 or PAM-8 with ADC-based RX require heavy parallel convolution units per aggressor that are considerably heavy in terms of area and power. Simulation results with one aggressor demonstrate that the MIMO OFDM can be effective for $>$ 200 Gb/s/lane dense electrical links beyond PAM.

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